

Harsath Kaliappa Gounder Thangavel

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Education

York University

Bachelor of Science (BSc) in Computer Science (Honours)

Toronto, Canada

January 2020 – December 2024

Professional Experience

Software Developer

Privex Inc.

October 2023 – January 2024

United Kingdom (Remote)

- Developed an AI-driven support assistant using Retrieval-Augmented Generation (RAG) on a self-hosted LLM, serving 800+ customers, reducing staff workload by handling inquiries related to support tickets, products and services.
- Implemented REST APIs to interface with the LLM and integrated Discord APIs for customer queries and staff data updates.
- Implemented Celery-Redis task queue to asynchronously process Discord LLM requests, preventing system overload.
- Built data ingestion and management workflows to process context data from support tickets, website documentation, bulk files, and Discord announcements.
- Built and maintained infrastructure using MySQL for chunked data storage and Chroma for embedding retrieval.
- Containerized the Flask API with Docker for deployment and configured systemd services with monitoring scripts.
- **Tech Stack:** Python (Flask, Chroma, Celery, Langchain), Redis, MySQL, Docker, Linux

Machine Learning Engineer

CogniPet AG

April 2019 – December 2019

Switzerland (Remote)

- Developed Pet Breed Detection system using Xception neural network and transfer learning, achieving 94% accuracy across 120 dog breeds.
- Implemented REST APIs for model inference and integrated them into CogniPet’s Android and iOS apps using AWS EC2.
- Built data management workflows for user-uploaded pet images in AWS S3 with GDPR compliance.
- Explored video-based techniques for pet facial recognition through multi-frame feature extraction analysis.
- **Tech Stack:** Python (Flask, TensorFlow, NumPy), AWS (EC2, S3)

Founder and Full-Stack Web Developer

TaffyHub.com

August 2017 – July 2019

Coimbatore, India

- Developed and launched school social networking platform, reaching 500+ registered users and 150+ daily active users.
- Implemented news feed, friend recommender, meme creation, engagement system, and premium membership features.
- Built backend with PHP and MySQL, frontend with JavaScript, jQuery, and Bootstrap, using Memcached for query optimization.
- Self-hosted the platform on Linux infrastructure, implementing reverse proxies, automated backups, and system maintenance as part of DevOps workflow.
- Generated \$100/month revenue through AdSense and premium memberships.
- **Tech Stack:** PHP, JavaScript, MySQL, Memcached, Linux (rsync, cron)

Projects

Jix Interpreter | C, CMake, GitHub Actions

- Developed an interpreter for a dynamically typed programming language in C.
- Implemented fault-tolerant parsing and error handling, arrays, function pointers, loops, nested blocks, recursion, built-in functions, conditional statements, and control flow statements.
- Implemented basic functional programming support, including first-class and higher-order functions.

Prism Coverage | Java, Maven, JUnit, GitHub Actions

- Developed a Java test code coverage engine to measure test suite coverage against control flow and data-flow criteria.
- Implemented color-coded HTML report generation highlighting covered and uncovered sections of source code.
- Built JUnit test suite and GitHub Actions CI/CD workflow to verify coverage engine functionality.

Technical Skills

Languages: Python, Java, JavaScript (ES6), TypeScript, C++ (C++17), C, Bash, HTML5/CSS3
Technologies: Linux, Docker, SQL (MySQL, Postgres), NoSQL (MongoDB, Redis, Neo4j), AWS
Frameworks: Node.js, React, Flask, Django, LangChain
Tools: Git, CI/CD (Github Actions, Jenkins), Build tools (CMake, Maven, Vite)